Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	2 October 2026
Team ID	
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The technical architecture presents the main layers of the solution and the way they are linked. Users come in through the portal, email and mobile. Their requests are handled by the ServiceNow Service Catalog and Flow Designer, which read and write data in platform tables and communicate with external systems such as the vendor.

Users	Employee	Manager	IT Procurement Coordinator	
Presentation layer	Employee Center / Service Portal	Email and mobile approvals	Reports and dashboards	
Application layer (ServiceNow)	Service catalog item: standard laptop	Flow Designer: approval flow	Flow Designer: fulfilment flow	Flow Designer: notification flow
Data layer	Request tables (sc_request, sc_req_item)	Asset inventory (alm_hardware)	Purchase orders (proc_po)	
Integrations	Vendor (REST API)	Identity provider (SSO)	Email server (SMTP)	

Flow Designer carries out the automation in the application layer: approvals, stock checks, purchase orders, asset records and notifications.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User interface	Employees request laptops and follow their status; managers and coordinators handle their assigned tasks	ServiceNow Employee Center / Service Portal
2	Service catalog	Standard laptop catalog item with pre-filled variables and eligibility checks	ServiceNow Service Catalog, Catalog Item
3	Workflow automation	Approval, fulfilment and notification flows that run with no manual steps	Flow Designer, Subflows, Actions
4	Approvals	Automatic approval for eligible roles, plus manager approval with escalation	Flow Designer approval action, Approval Engine
5	Database	Holds requests, assets and purchase orders	ServiceNow tables (sc_request, sc_req_item, alm_hardware, proc_po)
6	Notifications	Updates to the requester and approvers at every stage	ServiceNow Notifications, Email, Mobile
7	External integration	Sends purchase orders to the vendor that has been assigned	REST Message, IntegrationHub
8	Reporting	Dashboard for turnaround time and savings	ServiceNow Reports, Performance Analytics
9	Cloud infrastructure	Hosts the application and the database	ServiceNow cloud instance

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Low-code framework	Automation is built in a visual designer rather than with custom code	Flow Designer, ServiceNow Studio
2	Security implementations	Role-based access, single sign-on and encrypted connections	ACLs and roles, SAML SSO, HTTPS
3	Scalable architecture	New models, roles and vendors are added by configuration	Subflows, catalog variables, decision tables
4	Availability	Runs on a managed cloud platform	ServiceNow cloud infrastructure
5	Performance	Routine steps run automatically in the background, which cuts manual delay	Flow Designer asynchronous execution

Implementation Screenshots (ServiceNow)

Captured from the project demo video: Flow Designer build and a test laptop request.

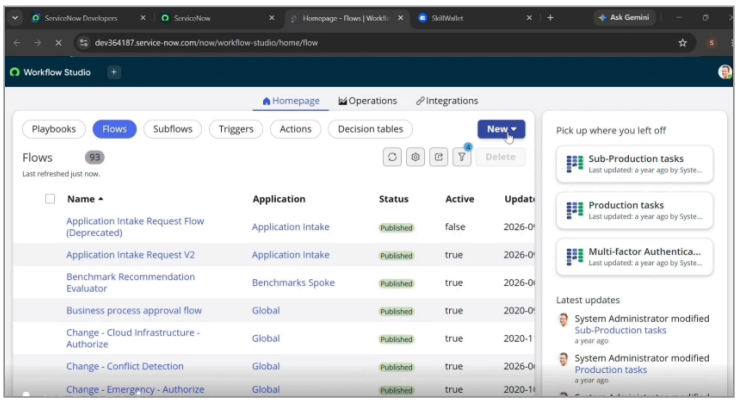


Figure 1: Workflow Studio home page in ServiceNow

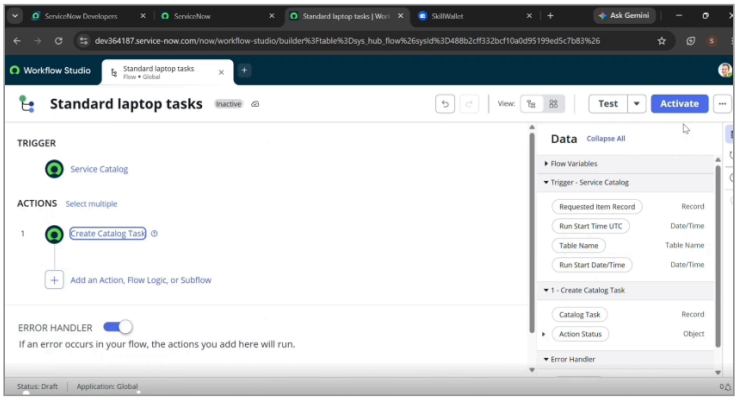


Figure 2: Flow Designer: Service Catalog trigger and Create Catalog Task action

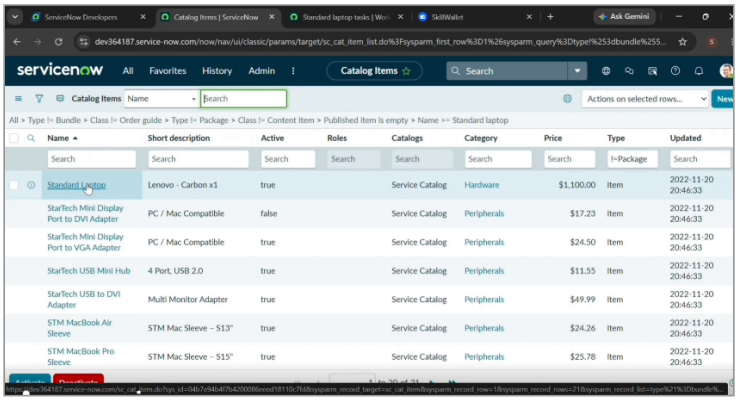


Figure 3: Catalog item "Standard Laptop" (Lenovo Carbon x1) in the catalog

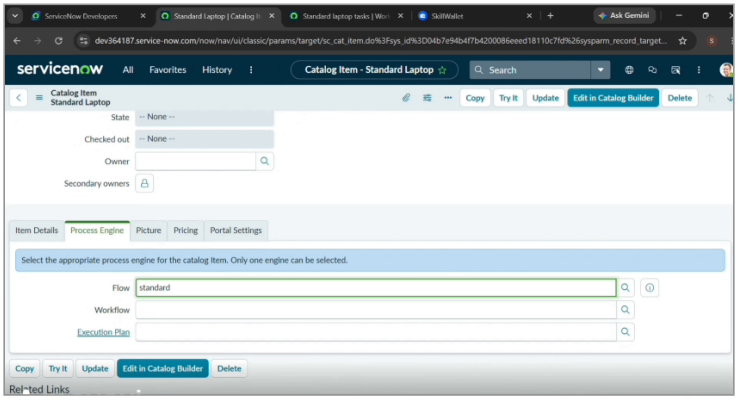


Figure 4: Catalog item Process Engine tab (fulfilment flow selection)